

30-Day Data Science & Machine Learning Roadmap

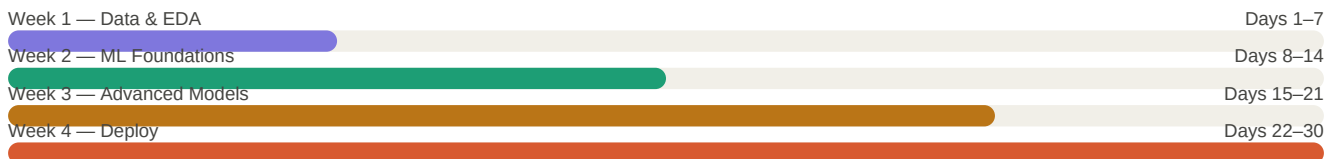
Intermediate level

1–2 hrs / day

30 days

4 weeks

YOUR JOURNEY AT A GLANCE



FINAL OUTCOMES AFTER 30 DAYS

Data fluency	Clean, explore and visualise any real-world dataset confidently
ML toolkit	Train, tune, and evaluate 6+ supervised & unsupervised models
Deep learning	Build and fine-tune neural networks with PyTorch or Keras
Portfolio	A live, deployed ML app you can show in interviews



Week 1 — Data Wrangling & Exploratory Analysis

Milestone: Clean & explore a real dataset

DAILY TASKS

Day 1	Review pandas basics: DataFrames, indexing, filtering, groupby
Day 2	Handle missing values & duplicates; explore <code>.info()</code> and <code>.describe()</code>
Day 3	Data types, type casting, datetime parsing, string operations
Day 4	Merge, join, and reshape datasets (pivot, melt, stack)
Day 5	Visualise distributions with matplotlib & seaborn (histograms, boxplots)
Day 6	Correlation analysis, heatmaps, pairplots; spot outliers
Day 7	EDA mini-project: full analysis on a Kaggle dataset, write findings

RESOURCES

- Kaggle pandas course — free, interactive, 4 hrs
- Python for Data Analysis (Wes McKinney) — chapters 1-6
- Seaborn gallery — seaborn.pydata.org/examples
- Titanic or House Prices dataset on Kaggle
- Corey Schafer pandas playlist on YouTube

PROJECT

Exploratory Data Analysis Report

Pick any dataset from Kaggle (e.g. World Happiness Report or Airbnb listings). Write a Jupyter notebook that profiles the data, cleans it, visualises 5 key patterns, and summarises 3 actionable insights. Export as HTML to share.

`pandas` · `seaborn` · `Jupyter`

Week 2

Week 2 — Core ML Algorithms

Milestone: Train & evaluate 3 models

DAILY TASKS

Day 8	Feature engineering: encoding, scaling, train/test split with sklearn
Day 9	Linear & logistic regression — theory + hands-on implementation
Day 10	Decision trees & Random Forests; understand overfitting vs underfitting
Day 11	Evaluation metrics: accuracy, precision, recall, F1, ROC-AUC, RMSE
Day 12	Cross-validation and hyperparameter tuning with GridSearchCV
Day 13	SVM and k-NN — when to use them, trade-offs
Day 14	Build an end-to-end classification pipeline; compare all 3 models

RESOURCES

- Hands-on ML with Scikit-Learn & TensorFlow (Aurélien Géron) — ch. 1-6
- fast.ai Practical ML for coders — free course
- scikit-learn documentation — tutorials & examples
- StatQuest with Josh Starmer — ML concept videos on YouTube

PROJECT

Binary Classifier: Loan Default or Churn Prediction

Use a credit or customer churn dataset. Build a full pipeline: preprocessing, 3 classifiers (logistic regression, random forest, SVM), evaluation with a confusion matrix and ROC curve. Write a 1-page model comparison report.

`scikit-learn` · `sklearn` pipelines · `matplotlib`

Week 3

Week 3 — Advanced Models & Deep Learning

Milestone: Train a neural network from scratch

DAILY TASKS

Day 15	Gradient Boosting: XGBoost & LightGBM — hyperparameters, early stopping
Day 16	Unsupervised learning: k-means clustering, PCA for dimensionality reduction
Day 17	Neural network fundamentals: perceptrons, activation functions, backprop
Day 18	Build a feedforward NN with PyTorch or Keras; train on tabular data
Day 19	CNNs: convolution, pooling; train on MNIST or CIFAR-10
Day 20	Transfer learning with a pretrained model (ResNet, EfficientNet) via Keras
Day 21	Experiment tracking with MLflow or W&B; log metrics, compare runs

RESOURCES

- [fast.ai Practical Deep Learning](#) — lessons 1-4 (free)
- [Deep Learning with Python](#) (François Chollet) — chapters 1-7
- [XGBoost & LightGBM docs](#) — parameter guides and examples
- [Andrej Karpathy's Neural Networks: Zero to Hero](#) (YouTube)
- [Weights & Biases quickstart](#) — [wandb.ai/quickstart](#)

PROJECT

Image Classifier with Transfer Learning

Pick a 2-5 class image dataset (dogs vs cats, disease detection). Fine-tune a pretrained EfficientNetB0 or ResNet50. Achieve >85% validation accuracy, log metrics to W&B, and document what happens when you freeze vs unfreeze layers.

[Keras](#) · [PyTorch](#) · [Transfer Learning](#) · [W&B](#);



Week 4 — Capstone Project & Deployment

Milestone: Ship a live ML web app

DAILY TASKS

Day 22	Choose capstone problem; collect & explore your dataset end-to-end
Day 23	Feature engineering and baseline model; set success metrics
Day 24	Model selection: compare 2-3 approaches, tune the winner
Day 25	Error analysis: where does the model fail? Add features or fix data
Day 26	Build a Streamlit or Gradio web app around your model
Day 27	Deploy to Hugging Face Spaces or Render; test it live
Day 28	Write a model card: problem, data, methods, results, limitations
Day 29	Polish the notebook and GitHub README; record a 2-min demo video
Day 30	Post on LinkedIn or Kaggle; reflect on what to learn next

RESOURCES

- Streamlit docs — docs.streamlit.io, build an app in 30 min
- Hugging Face Spaces — free deployment for ML apps
- Model cards for model reporting — Mitchell et al. (2019)
- Made With ML — madewithml.com, MLOps overview

PROJECT

Capstone: End-to-End ML Application

Ideas: sentiment analyser, house price predictor, medical image classifier, or customer churn dashboard. Must include a live demo URL, a clean GitHub repo with README, a model card, and at least one visualisation of model performance.

[Streamlit](#) · [Gradio](#) · [Hugging Face](#) · [GitHub](#)

Daily Study Schedule

60-min day (minimum)

On busy days — never skip, just scale down

Time	Phase	Activity
0:00 – 0:10	Warm-up	Review yesterday's notes or code for 10 min
0:10 – 0:40	Concept	Read or watch the day's core concept
0:40 – 0:55	Code	Write at least 20 lines — reproduce one example
0:55 – 1:00	Note	Write 2 sentences: what did you learn, what confused you?

90-min day (target)

The ideal session — enough to make real progress

Time	Phase	Activity
0:00 – 0:10	Warm-up	Review yesterday. Fix one thing that was unclear
0:10 – 0:35	Concept	Read / watch the core concept. Take notes
0:35 – 1:10	Code	Implement from scratch on today's dataset
1:10 – 1:20	Extend	Try one variation — different parameter or chart
1:20 – 1:30	Note	Commit code to GitHub. Write today's reflection

120-min day (deep work)

For project days (Day 7, 14, 21, 22-30)

Time	Phase	Activity
0:00 – 0:10	Warm-up	Review week's progress. Set one clear goal
0:10 – 0:30	Concept	Fill any gaps with reading or targeted search
0:30 – 1:30	Build	Uninterrupted project work — no phone, no tabs
1:30 – 1:50	Review	Code review: clean up, add comments, push to GitHub
1:50 – 2:00	Reflect	Write a short project log entry. What's left to do?

WEEKLY RHYTHM RULES

- Monday–Friday: concept + code (60–90 min)
- Saturday: project day (90–120 min, build something)
- Sunday: review + plan (30 min only — read notes, plan next week)

- Never miss two days in a row. One skip is rest; two is a reset.
- Keep a learning-log.md in your GitHub repo — one entry per day